

Daniel Hathcock

PHD CANDIDATE IN ACO · CARNEGIE MELLON UNIVERSITY, PITTSBURGH, PA

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Education

Carnegie Mellon University

Pittsburgh, PA

PH.D. CANDIDATE IN ALGORITHMS, COMBINATORICS, AND OPTIMIZATION (ACO)

Aug. 2020 - May 2026

- Advised by Professor R. Ravi.
- Research in approximation algorithms for network design and resource allocation.
- NSF Graduate Research Fellowship recipient.

Carnegie Mellon University

Pittsburgh, PA

M.S. IN ALGORITHMS, COMBINATORICS, AND OPTIMIZATION (ACO)

Aug. 2020 - Dec 2023

- Awarded after completion of Ph.D. course requirements

Georgia Institute of Technology

Atlanta, GA

B.S. IN COMPUTER SCIENCE, MINOR IN MATHEMATICS

Aug. 2016 - May 2020

- Concentrations in Theory and Intelligence.
- Undergraduate Thesis: Enumerating Acyclic Orientations, advised by Professor Prasad Tetali

Research Publications

Perfect Fractional Matchings in Bipartite Graphs Via Proportional Allocations

preprint

WITH R. RAVI

2025

CoRR preprint, 2025. arxiv.org/pdf/2510.01107

The Telephone k -Multicast Problem

Algorithmica

WITH GUY KORTSARZ, R. RAVI

2026

Algorithmica, Volume 88, article number 25, (2026). doi.org/10.1007/s00453-026-01377-5

The Steiner Path Aggregation Problem

IPL vol.192

WITH DA QI CHEN, D ELLIS HERSHKOWITZ, R. RAVI

2026

Information Processing Letters, vol. 192:106608, 2026. arxiv.org/pdf/2510.01392

The Online Submodular Assignment Problem

FOCS'24

WITH BILLY JIN, KALEN PATTON, SHERRY SARKAR, MIK ZLATIN

2024

IEEE Symposium on Foundations of Computer Science (FOCS) 2024. arxiv.org/pdf/2401.06981

Approximation Algorithms for Steiner Connectivity Augmentation

ESA'24

WITH MIK ZLATIN

2024

LIPICs, Volume 308, European Symposium on Algorithms (ESA) 2024. arxiv.org/pdf/2308.08690

ESA best student paper award

The Telephone k -Multicast Problem

APPROX'24

WITH GUY KORTSARZ, R. RAVI

2024

LIPICs, Volume 317, APPROX/RANDOM 2024. arxiv.org/pdf/2410.01048

Maintaining Matroid Intersections Online

SODA'24

WITH NIV BUCHBINDER, ANUPAM GUPTA, ANNA R. KARLIN, SHERRY SARKAR

2024

ACM-SIAM Symposium on Discrete Algorithms (SODA24). arxiv.org/pdf/2309.10214

One Tree to Rule Them All: Poly-Logarithmic Universal Steiner Tree

FOCS'23

WITH COSTAS BUSCH, DA QI CHEN, ARNOLD FILTSEY, D ELLIS HERSHKOWITZ, RAJMOHAN RAJARAMAN

2023

IEEE Symposium on Foundations of Computer Science (FOCS) 2023. arxiv.org/pdf/2308.01199

Toppleable Permutations, Excedances and Acyclic Orientations

CT-2(1)

WITH ARVIND AYYER, PRASAD TETALI

2022

Combinatorial Theory, 2(1). arxiv.org/pdf/2010.11236

Professional Experience

TEACHING AND MENTORING

Fall 2024	Network Optimization I (47-835) , Grader	CMU
Fall 2023	Network Optimization I (47-835) , Grader	CMU
Summer 2023	Polymath Jr. Research Program , Graduate Mentor for Covering Grids with Hyperplanes Project	Polymath Jr.
Summer 2022	Polymath Jr. Research Program , Graduate Mentor for Ramsey Theory Project	Polymath Jr.
Summer 2021	Concepts of Mathematics (21-127) , Graduate Teaching Assistant	CMU
Spring 2021	Discrete Mathematics (21-228) , Graduate Teaching Assistant	CMU
Fall 2020	Calculus in 3 Dimension (21-259) , Graduate Teaching Assistant	CMU
Fa. '19 - Sp. '20	Design & Analysis of Algorithms (CS 3510) , Teaching Assistant	Georgia Tech
Fall 2018	Intro to Linear Algebra (Math 1553) , Teaching Assistant	Georgia Tech

INTERNSHIPS

Summer 2020	Tagup, Inc. , Data Science Intern	Somerville, MA
	Developed and implemented machine learning models using JAX and Tensorflow for survival analysis, predicting time-to-event (TTE). Enabled >10x speedup of distributed model training using Dask.	
Summer 2017	Left Hand Robotics , Prototyping / Software Engineering Intern	Longmont, CO
	Worked on the design and control of a self-driving snow removal robot. Prototyped algorithms for high precision GPS path collection, following, and correction. Used Python and Java being run on a Raspberry Pi.	

PROFESSIONAL SERVICE

2023	CMU Theory Lunch Seminar , Co-organizer	CMU
Fa. '19 - Sp. '20	Georgia Tech Theory CS Club , Talk Coordinator	Georgia Tech

Honors & Awards

ACADEMIC

2024	ESA best student paper award , for 'Approximation Algorithms for Steiner Connectivity Augmentation'	London, UK
2021	NSF Graduate Research Fellowship Program , Awarded (3 years PhD funding)	NSF
2020	Highest Honor , Georgia Tech institutional honors	Atlanta, GA
2016-2020	Faculty Honors Letter , 4.0 GPA all semesters at Georgia Tech	Atlanta, GA

PROFESSIONAL

2022	CMU ACM Hackathon: Algorithms with a Purpose , Second Place	Pittsburgh, PA
2021	CMU ACM Hackathon: Algorithms with a Purpose , Second Place	Pittsburgh, PA
2018	HackGT: Goldman Sachs Data Mining Challenge , First Place	Atlanta, GA
2017	HackGT: FINRA Data Mining Challenge , First Place	Atlanta, GA

Misc

Languages	Python/SageMath, Mathematica, Java, C/C++, $\text{\LaTeX}$
Skills/Tools	Scientific Python (SciPy, NumPy, scikit, Jupyter, etc.), Gurobi, OR-Tools, Git. Familiarity with: TensorFlow, PyTorch, Dask, OpenCV
Service	Math Tutoring at Minadeo Elementary, a Title I school in Pittsburgh, PA (2022)